Hamiltonian Boundary Value Method HBVM(4,2)

Rank-two polynomial path with four Gauss nodes

1 Polynomial-path formulation

Let $(c_i, b_i)_{i=1}^4$ be four-point Gauss–Legendre quadrature on $[0, 1]$. For the first two orthonormal shifted Legendre polynomials,

$$P_0(c) = 1, \qquad P_1(c) = \sqrt{3}(2c - 1),$$

define

$$I_{i0} = c_i, \qquad I_{i1} = \sqrt{3}(c_i^2 - c_i).$$

HBVM(4,2) solves for two vector coefficients γ_0, γ_1:

$$Y_i = y_n + h \sum_{j=0}^{1} I_{ij}\gamma_j, \tag{1}$$

$$R_j(\gamma) = \gamma_j - \sum_{i=1}^{4} b_i P_j(c_i) f(t_n + c_i h, Y_i) = 0. \tag{2}$$

The accepted state is

$$y_{n+1} = y_n + h \sum_{i=1}^{4} b_i f(t_n + c_i h, Y_i).$$

Equivalently, the Runge–Kutta matrix is $A = \mathcal{I}_2 \mathcal{P}_2^T \Omega_q$ and has rank two. The nonlinear unknown therefore has twice the state dimension, although four stages evaluate the line integral.

2 Order and energy property

With polynomial degree $s = 2$, the method has classical order $2s = 4$. Four-point Gauss quadrature has degree seven. For an autonomous polynomial Hamiltonian of degree ν, exact polynomial energy conservation follows when $k \geq \nu s/2$; hence HBVM(4,2) covers degrees through four. For a general smooth Hamiltonian the quadrature error makes energy conservation approximate, with an error that decreases rapidly with the step and quadrature accuracy.

HBVM(k, s) is generally not symplectic when $k > s$. Thus HBVM(4,2) must not be labeled symplectic by construction; map-Jacobian diagnostics measure its canonical defect independently of its energy behavior.

3 Nonlinear configurations

Newton acts on the reduced residual (R_0, R_1). Pass `solver=NewtonSolver(...)`; its `jacobian_method` selects differentiation. `analytic` uses the dense or particle-block Jacobian supplied by execution; `finite_difference` differentiates stage fields; `auto` uses the supplied derivative when available. The solver owns tolerances and iteration limits. HBVM defaults to `backtracking=True`, which shortens corrections when the full Newton update increases the residual. The stopping scale is

$$\varepsilon_{\text{abs}} + \varepsilon_{\text{rel}} \max(1, \|y_n\|_\infty).$$

The generic `EnergyObserver` evaluates physical H and analytic positive $\partial_t H$ at accepted endpoints and requested physical samples. It records energy differences without integrating momentum or

using internal stages. A historical $H + \kappa$ study must explicitly request the separate stage-based diagnostic. Physical trajectories and solver equations are unchanged.

4 Implementation correspondence

Coefficients, residuals, Newton assembly, and integration are in `src/methods/hbvm/order4.py`. The simulation document describes required Hamiltonian capabilities, work counters, fixed-grid sampling, and evaluation studies; shared physical dynamics are documented outside the model tree.

5 Physical state and endpoint energy observations

Accepted states contain one physical vector $Z \in \mathbb{R}^{dN}$ and scalar time. GC vectors have $2N$ entries and FC vectors have $4N$; doubled workspaces are transient. InitialConditions supplies values and layout during preparation only. The numerical base retains functions, metadata, run status and copy count; execution owns the initial snapshot and output schedule. Supplied Jacobians are independent particle blocks (N, d, d); finite differences use dense matrices.

The generic `EnergyObserver` evaluates only

$$H_n = H(t_n, Z_n), \quad H_{n+1} = H(t_{n+1}, Z_{n+1}), \quad (\partial_t H)_n, \quad (\partial_t H)_{n+1}, \quad \Delta H_n = H_{n+1} - H_n.$$

It also evaluates these physical functions at requested saved samples, without consuming internal stages or integrating a momentum coordinate. The derivative is positive, analytic and partial at fixed state. GC differentiates its normalized Hamiltonian potential with `dt=1`; spatial splines and Fourier time factors retain their original scaling. This is not the secant $\Delta H_n/h$. Variation of a time-dependent Hamiltonian is not a numerical conservation error.

Specialized historical generalized-energy studies separately reconstruct $\dot{\kappa} = -\partial_t H$ from retained numerical stages. Their signed weights, physical normalization and archived diagnostics keep their original meaning. They are not part of generic energy observation. Host and JAX execution share the fixed numerical map and physical-only accepted state. JAX evaluates endpoint energy on the device without a momentum accumulator.

6 Optional JAX execution

The optional JAX driver executes the same map in double precision on one selected CPU or GPU, batching particles inside a compiled sequential time loop. It preserves coefficients, physical tolerance scaling, signed stage times, passive quadrature, independent shortened output maps and convergence failures. Newton uses the supplied particle-block structure. Both routes return the same physical solution layout; floating-point reassociation can change round-off without changing the designed order or geometric properties. First-call compilation is separate from warmed timings, which include transfers and output. The shared execution contract specifies callbacks and device scope.